

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Load a static image as cv2.VideoCapture(0) is not su
image_path = "/content/wp3892792.jpg"
img = cv2.imread(image_path)

if img is None:
    print(f"Error: Could not load image from {image_pa
else:
    # Define the region of interest points in the sour
    roi_points = np.array([(150, 200), (450, 200), (500, 600), (150, 600)])

    # Define the corresponding points in the target im
    target_points = np.array([(0, 0), (400, 0), (400, 600), (0, 600)])

    # Get the perspective transformation matrix
    M = cv2.getPerspectiveTransform(roi_points, target

    # Apply the perspective transformation
    output_width = 400
    output_height = 600
    dst = cv2.warpPerspective(img, M, (output_width, o

    # Convert the transformed image from BGR to RGB fo
    dst_rgb = cv2.cvtColor(dst, cv2.COLOR_BGR2RGB)

    # Display the transformed image using Matplotlib
    plt.figure(figsize=(output_width / 100, output_he
    plt.imshow(dst_rgb)
    plt.title('Perspective Transformed Image')
    plt.axis('off')
    plt.show()

    # Optionally save the transformed image
```

```
cv2.imwrite('Perspective_Transformed_Image_Output.jpg', transformed_image)
print("Image 'Perspective_Transformed_Image_Output.jpg' saved successfully.")
```

Perspective Transformed Image



Image 'Perspective_Transformed_Image_Output.jpg' saved successfully.

